

replace conventional vitrified-bond aluminum-oxide grinding wheel with a green silicon carbide wheel

Executive Technology Brief

Domain: mechanical_structural

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented manufacturing problem: when grinding titanium alloys (Ti-6Al-4V) with conventional aluminum-oxide wheels, titanium chips adhere to the wheel's abrasive grains and bond (wheel loading); the loaded wheel stops cutting, friction rises, the wheel glazes, and the workpiece surface temperature spikes — producing tensile residual stresses, alpha-case burn, microcracking, and scrapped expensive aerospace parts. Titanium's low thermal conductivity and chemical reactivity with abrasive make this worse than steel grinding. Qualitative engineering-literature context; no fabricated statistics (retrieval must custody the evidence).

Mechanism: green silicon carbide wheel maintained better thermal stability and wear resistance

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: replace conventional vitrified-bond aluminum-oxide grinding wheel with a green silicon carbide wheel

Governing Model: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention:

APPLICABLE; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design. Engaged variables: σ .

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: APPLICABLE; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead. Engaged variables: σ .

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.